Alexandros **Alexiou**

Senior Software Engineer · Backend & Distributed Systems

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Summary

Senior Backend Engineer and end-to-end problem solver with 5+ years building microservices and event-driven systems in Kotlin/Java and C++, owning work from design to production. At BestSecret, I build high-traffic services on Kafka, PostgreSQL and Kubernetes, from breaking the Product Detail Page out of the monolith and designing GraphQL APIs for mobile to an AI-driven visual recommendation platform and customer notification pipelines handling ~2M messages/day. Experienced in zero-downtime migrations and performance tuning for flash-sale peaks.

Work Experience

Senior Software Engineer

Ioannina, Greece - Remote

BestSecret GmbH — *Dec. 2023 - present*

- Redesigned the Low Stock Notification pipeline (~2M notifications/day), cutting end-to-end processing from hours to minutes; added wishlist-based targeting with per-customer and global caps, consistently outperforming the legacy system on attributed visits and orders across all attribution windows.
- Built the Back In Stock service end-to-end (Kotlin, Spring Boot, Kafka, PostgreSQL), dispatching ~1.5M notifications/month: idempotent Kafka consumers for stock-state transitions, tier-based scheduling via Salesforce Marketing Cloud, and retry-with-muting for delivery resilience; exposed subscriptions to web and mobile through REST and GraphQL APIs.
- Migrated Back In Stock off the legacy system with zero downtime: shadow-mode scheduler alongside the old one, dual-writes, a backfill tool, and idempotent upserts with Kafka-backed retry to tolerate eventual consistency.
- Designed an event-driven visual recommendation platform on AKS (Spring Boot, Kafka, ONNX Runtime/CLIP image embeddings, pgvector similarity search) that asynchronously processes new product images and generates visually matched set recommendations.
- Decomposed the Product Detail Page (PDP) from the monolith into a Kotlin/Spring Boot microservice (Kafka, PostgreSQL) and an Astro.js/Vue.js/TypeScript frontend; designed a GraphQL API for the mobile apps that reduces over-fetching; delivered with full unit and E2E test coverage alongside cross-functional teams.
- Built a high-throughput Kafka consumer that processes and persists records to cloud storage, tuned for throughput and data consistency.
- Load-tested critical services with k6 ahead of flash-sale peaks; introduced Redis caching on PDP product reads with Kafka-driven invalidation and a PostgreSQL read replica for the wishlist service, cutting primary DB load and latency under peak traffic.
- Set up Datadog monitoring and alerting, built unit and integration test suites with JUnit and Testcontainers across owned services, and regularly review merge requests across the team.

Software Engineer (Part-time)

Ioannina, Greece - Remote

Telepost – Pneumatic Tube Systems — *Mar. 2025 - Jul. 2026*

- On-call for live customer installations: diagnosed and recovered production failures (stuck transmissions, jammed routing, power-outage recovery) and fed recurring failure patterns back into product hardening and test coverage.
- Migrated historical statistics from per-date text files to an indexed SQLite database, eliminating multi-minute load times and enabling fast date-range queries and aggregations; the migration was idempotent and reported unparseable legacy rows so no data was silently lost.
- Led a CAD-style redesign of the C++17 schematic editor (wxWidgets, SFML); shipped per-station predictive-maintenance limits.

Software Engineer (promoted from Intern and Junior)

Ioannina, Greece - Hybrid

TeamViewer Germany GmbH — *Oct. 2021 - Dec. 2023*

- Built a Java MongoDB-to-SQL Server migration and sync pipeline for business-critical IoT data from medical centers and enterprise customers, with zero data loss during cutover: outbox-based change capture replayed continuously to SQL, a separate backfill service for historical data, chunked processing, idempotent writes, exponential-backoff retries, and persistent failure tracking and replay.
- Extended TeamViewer's public enterprise SaaS Web API by integrating new functionality into C++17 microservices, exposing scripting previously limited to the Management Console UI; on own initiative, extended the open-source TeamViewerPS PowerShell module accordingly.
- Integrated an acquired IoT product into TeamViewer's architecture, porting its Java microservices and Node.js gateway to a C#/.NET service layer integrated with native C++ components via C++/CLI; worked directly with medical-center and enterprise customers alongside Support to gather requirements and diagnose field issues.
- Maintained core C++ backend services and their C# integrations over a custom binary serialization protocol; diagnosed production issues across service logic and protocol handling and delivered targeted fixes.
- Contributed to the TeamViewer Remote migration to a single TypeScript UI for desktop (WebView2) and web: implemented HTTP request handlers in the native C++ client and OpenAPI contracts on both platforms to port legacy features, and built the client restart flow with Windows Registry persistence to switch users between the old and new UI.
- Developed the WebAssembly browser client (C++/Emscripten, React.js, TypeScript), working on multithreaded C++ compiled to Web Workers so heavy processing runs off the browser's main thread.

Skills

Languages	Kotlin · Java · C++17 · C#/.NET · TypeScript/JavaScript · Python
Backend	Spring Boot · Microservices · Event-Driven Architecture · Kafka · REST APIs · GraphQL · Node.js · SOLID · Design Patterns
Data	PostgreSQL (pgvector, read replicas) · MongoDB · Microsoft SQL Server · Redis · Caffeine
Cloud & DevOps	Microsoft Azure (AKS) · Kubernetes · Helm · Docker · GitLab CI/CD · Git
Quality & Ops	Datadog · k6 · JUnit · Testcontainers · Cypress
Frontend	React · Vue.js · Astro.js
Spoken	English (Proficient) · Greek (Native)

Education

M.Eng. in Computer Science & Engineering *Ioannina, Greece*

University of Ioannina – Polytechnic School *Sept. 2015 - Mar. 2022*

- Thesis: Java library that automatically profiles an input dataset and generates a statistical report of the findings.